

Theoretical Analysis of Guided Mode and Effective Refractive Index at 1.53 μm in Ti:LiNbO₃ Strip Waveguides

Delong Zhang, Guilan Ding, and Caihe Chen

Abstract—An analysis of guided modes and effective refractive indexes in *c*-cut Ti-diffused LiNbO₃ strip waveguides has been performed at the wavelength 1.53 μm by the variational method. Dependences of mode size and effective refractive index on initial Ti-strip width W , diffusion temperature T and initial Ti-strip thickness H have been calculated for several lower order modes. As compared the plots of the mode size with the previous published experimental results, an agreement in general behavior has been obtained. The single-mode and cutoff conditions were discussed numerically, which are very useful in optimizing and designing the Er:Ti:LiNbO₃ waveguide devices. In addition, several advantages of 1.48 μm pumping were summarized.

Index Terms—Cutoff condition, diffusion parameters, effective refractive index, Ti:LiNbO₃ waveguide.

I. INTRODUCTION

ER-DOPED LiNbO₃ has been extensively studied as a promising material that combines the excellent laser properties of Er³⁺ ions with the electrooptic, acoustooptic, and nonlinear optical properties of LiNbO₃. In particular, it allows broadband amplification and lasing in the important wavelength range around 1.53 μm , which is preferred in future communication systems due to minimum fiber losses. In addition, this doping could significantly increase the potential of this material allowing a monolithic integration of active and passive components. To date, much work has been done, and includes mainly following several aspects.

- 1) The successful demonstration of active and passive devices, e.g., CW-emitting, free running waveguide laser [1]–[4]; also mode-locked [5], [6], tunable [7], and DBR waveguide lasers [8].
- 2) The work mentioned above is accompanied by quantitative modeling of the laser devices to improve the understanding of the physical phenomena involved and to allow an optimization of the laser properties [3], [9]–[14].
- 3) The investigation of the spectroscopic properties of Er³⁺ in Ti-diffused Er:LiNbO₃ waveguides [15].

- 4) A large amounts of theoretical and experimental studies have been performed for Ti:LiNbO₃ strip waveguides on mode dispersion by the effective index method [16], Ti-induced refractive index change as a function of diffusion parameters [17], optical properties [18], guided mode field distribution [19]–[21], etc. Of which the effective refractive index or the propagation constant, optical properties including mode cutoff condition, single-mode condition and waveguide losses, and the lower order modes have been studied mainly at 0.6, 1.15, and 1.32 μm wavelengths. At 1.53 μm , only the studies of mode field distributions were presented by effective refractive index method (ERI) [10], finite element method (FEM) and fast Fourier transform beam propagation method (FFT-BPM) [21].

In this paper, an analysis of guided mode and effective refractive index in Ti:LiNbO₃ waveguides has been carried out around 1.53 μm using variational method. The mode sizes and the effective refractive indexes as function of the controllable fabrication parameters such as initial Ti-strip width, diffusion temperature and initial Ti strip thickness were numerically calculated not only for fundamental mode, which is commonly encountered in actual lasers, but also for lower order modes, then, mode cutoff condition and single-mode condition were outlined.

II. CALCULATED RESULTS AND DISCUSSIONS

A. Theoretical Description

The variational analysis of Ti:LiNbO₃ waveguides provides an effective approach in determining simultaneously the mode sizes and the propagation constant or the effective refractive index [13]–[14], [19], [22], which are potentially useful in optimizing and designing waveguide devices. As compared with the extended effective index method and effective index method, in addition to the characteristic described above, it also shows speediness in calculation, simplicity in mathematical style and high accuracy in both propagation constant and the dominant field distribution of fundamental and lower order modes. According to Marcuse [23], the arbitrary order mode field distribution can be expressed as

$$\phi(x, y) = \frac{2}{\sqrt{W_x W_y}} \psi_n \left(\frac{\sqrt{2}x}{W_x} \right) \psi_{2m+1} \left(\frac{\sqrt{2}y}{W_y} \right) \quad (1)$$

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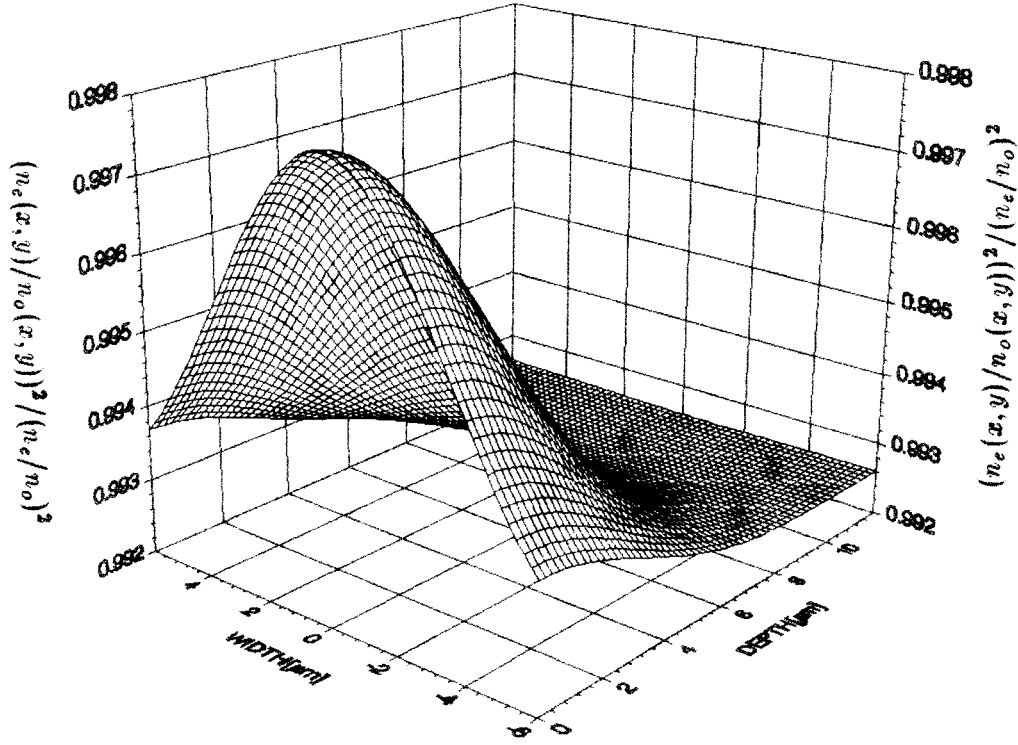


Fig. 1. Surface plot of the ratio $(n_e(x, y)/n_o(x, y))^2 / (n_e/n_o)^2$.

where

$$\psi_j(\xi) = \frac{1}{\sqrt{2^j j! \sqrt{\pi}}} \exp\left(-\frac{\xi^2}{2}\right) H_j(\xi) \quad (2)$$

$H_j(\xi)$ is the n th-order Hermite polynomial, n and $2m + 1$ are the modes orders in the x and y directions, respectively, corresponding to the width and depth directions of the guide. W_x and W_y are the mode sizes. Generally, only the lowest-order guided mode in the depth direction is considered, i.e., $m = 0$.

For TM modes guided in a c -cut Ti-diffused LiNbO₃ waveguide, a scalar, two-dimensional (2-D) wave equation for $\phi(x, y)$, according to Strake [20], can be written as

$$\left(\frac{n_e(x, y)}{n_o(x, y)}\right)^2 \nabla^2 \phi(x, y) + k_0^2 n_e^2(x, y) \phi(x, y) = \beta^2 \phi(x, y) \quad (3)$$

in which k_0 is the wavenumber in free space, $n_e(x, y)$ and $n_o(x, y)$ are the extraordinary and ordinary refractive index profiles, respectively, and are given by [16]

$$n_{e,o}^2(x, y) = \begin{cases} n_{eb,ob}^2 + (n_{es,os}^2 - n_{eb,ob}^2) f\left(\frac{y}{d_{ye,o}}\right) g\left(\frac{2x}{W}\right) & y > 0 \\ n_{eb,ob}^2 & y < 0 \end{cases} \quad (4)$$

where

$$f\left(\frac{y}{d_{ye,o}}\right) = \exp\left(-\frac{y^2}{d_{ye,o}^2}\right)$$

$$g\left(\frac{2x}{W}\right) = \frac{1}{2} \left[\operatorname{erf}\left(\frac{W}{2d_{xe,o}} \left(1 + \frac{2x}{W}\right)\right) + \operatorname{erf}\left(\frac{W}{2d_{xe,o}} \left(1 - \frac{2x}{W}\right)\right) \right] \quad (6)$$

To simplify equation (3), in Fig. 1 we show a surface plot of the ratio of $(n_e(x, y)/n_o(x, y))^2 / (n_e/n_o)^2$ at wavelength 1.53 μm (for the waveguide fabricated by indiffusing 7 μm width, 100 nm thickness Ti-strip at 1050 $^\circ\text{C}$ during 8.3333 h). It can be seen from Fig. 1 that $(n_e(x, y)/n_o(x, y))^2 / (n_e/n_o)^2 \sim 1$, i.e., $(n_e(x, y)/n_o(x, y))^2 \sim (n_e/n_o)^2$, then we approximate (3) as follows:

$$\left(\frac{n_e}{n_o}\right)^2 \nabla^2 \phi(x, y) + k_0^2 n_e^2(x, y) \phi(x, y) = \beta^2 \phi(x, y). \quad (7)$$

Inserting (1), (2), and (4) into (7), one can find the following variational equation for the propagation constant:

$$\beta^2 = k_0^2 n_{eb}^2 + \frac{2\sqrt{2}(n_{es}^2 - n_{eb}^2)k_0^2}{W_x} \left(1 + \frac{W_y^2}{2d_{ye}^2}\right)^{-\frac{3}{2}} Q - \left(\frac{n_e}{n_o}\right)^2 \frac{2n+1}{W_x^2} - \left(\frac{n_e}{n_o}\right)^2 \frac{3}{W_y^2} \quad (8)$$

where

$$Q = \int_0^{+\infty} g\left(\frac{2x}{W}\right) \psi_n^2\left(\frac{\sqrt{2}x}{W_x}\right) dx. \quad (9)$$

(5) If the optimum values of W_x and W_y correspond to the largest value of β^2 , namely β_m^2 , then the effective refractive index can

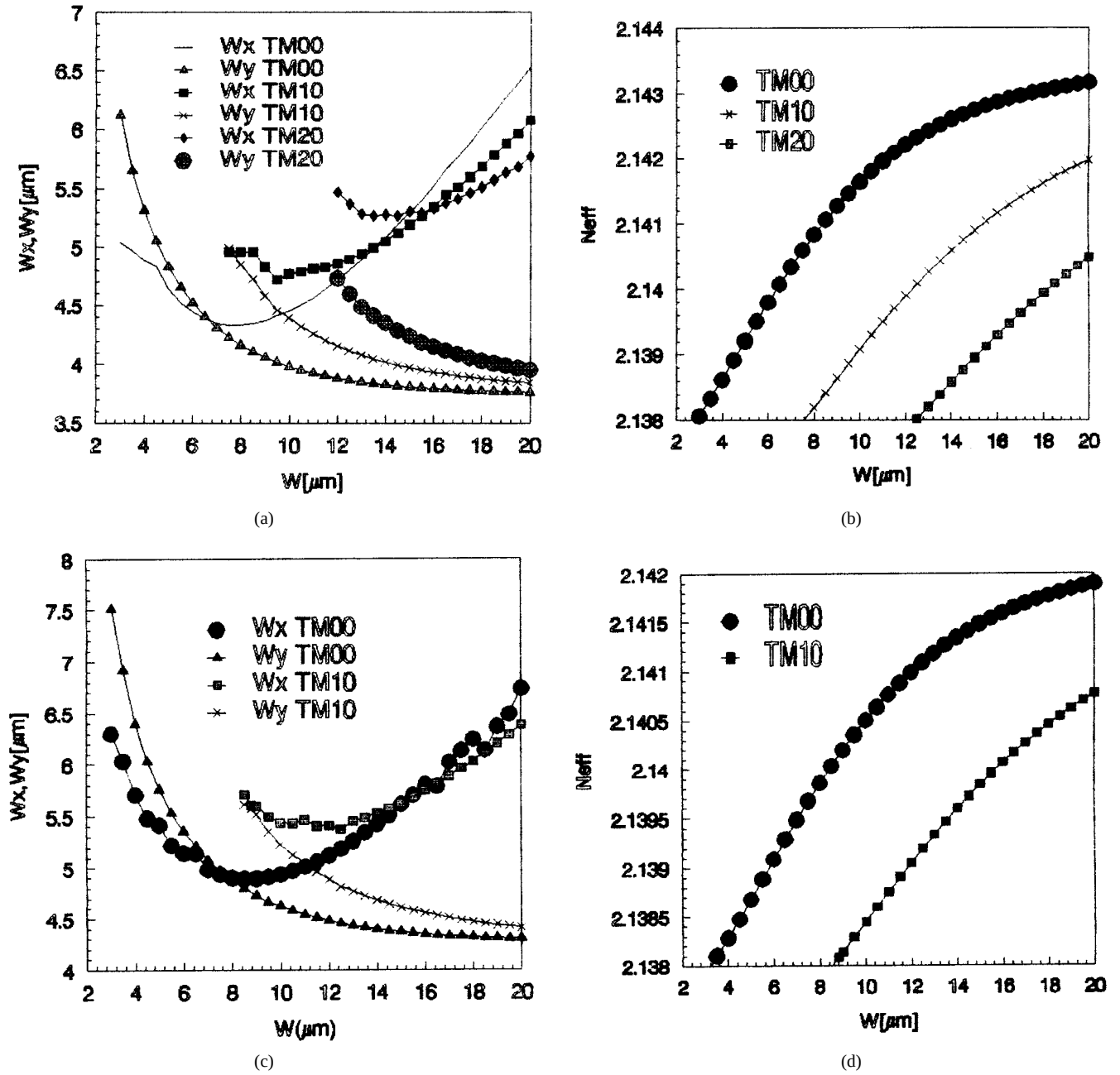


Fig. 2. Mode sizes and effective refractive indexes versus initial Ti strip width W . (a) and (b) for the waveguides fabricated by indiffusing 100 nm thickness Ti-strip at 1050 °C during 8.3333 h, (c) and (d) diffusion parameters taken from experimental data in [15].

be determined from the following expression:

$$N_{\text{eff}} = \frac{\lambda}{2\pi} \beta_m. \quad (10)$$

One can see from (8) that the propagation constant (or the effective refractive index), for a given wavelength, is determined mainly by the difference between two competing terms, one term depends on the overlap of the mode intensity distribution with the index profile [i.e., the second term in (8)], and the other term depends inversely on the square of the mode size [the latter two terms in (8)].

The concerned parameters and detailed calculation procedure have been described in the previous paper [13]. In the following we restrict our discussion to a c -cut Ti-diffused

LiNbO₃ waveguide and the calculations were performed for TM_{n0} modes at 1.53 μm .

B. Strip Width Dependence

Fig. 2(a) and (b) shows mode sizes of TM_{00} (fundamental mode), TM_{10} , and TM_{20} and the corresponding effective refractive indexes N_{eff} as function of initial Ti-strip width W . The other variables are commonly used values: diffusion temperature $T = 1050$ °C, initial Ti strip thickness $H = 100$ nm and diffusion time $t = 8.3333$ h. It is obvious from Fig. 2(a) that the waveguide with 3–7.5 μm wide Ti strip supports only the fundamental modes. Below 3 μm , the mode is expected to cutoff, and this is based on the fact

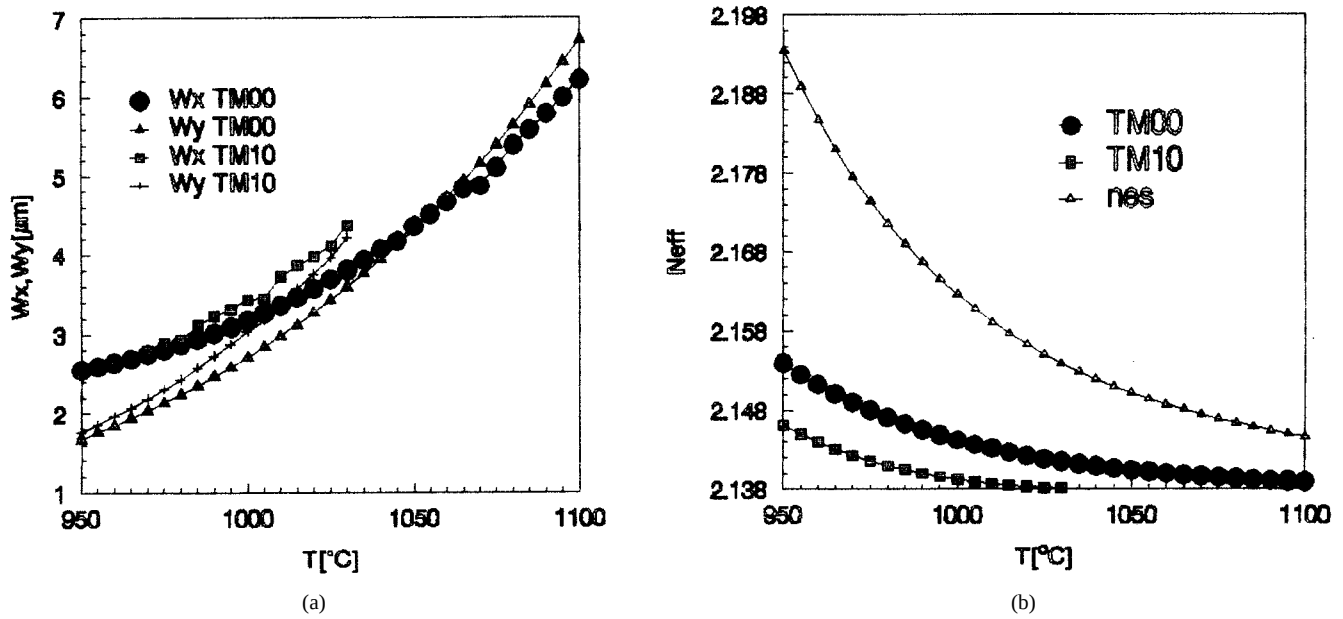


Fig. 3. (a) Mode sizes versus diffusion temperature for the waveguides fabricated by indiffusing $7 \mu\text{m}$ width, 100 nm thickness Ti-strip during 8.3333 h . (b) Corresponding effective refractive indexes.

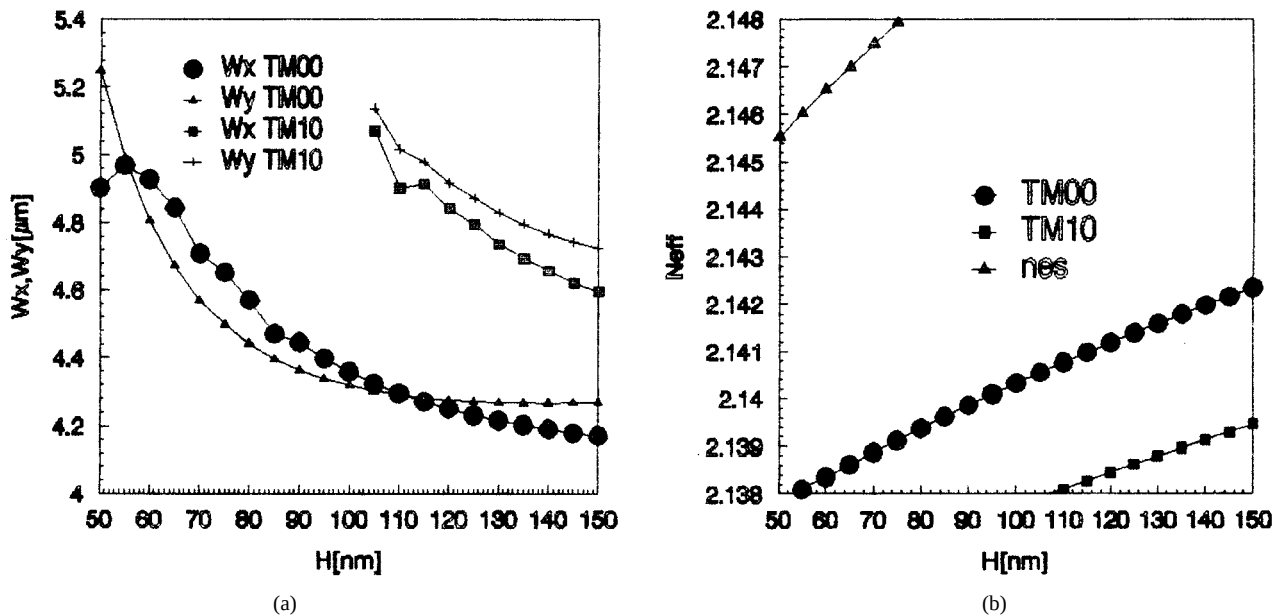


Fig. 4. (a) Mode sizes versus initial Ti-strip thickness H for the waveguides fabricated by indiffusing $7 \mu\text{m}$ width Ti-strip at $1050 ^{\circ}\text{C}$ during 8.3333 h . (b) Corresponding effective refractive indexes.

$n_{\text{eb}} < N_{\text{eff}} < n_{\text{es}}$ (here substrate index $n_{\text{eb}} = 2.1380$ and surface index of the guide $n_{\text{es}} = 2.1502$). When the stripe width W is increased to $7.5 \mu\text{m}$, the multimode region is approached. In the single-mode region mentioned above, both W_x and W_y decrease with W , and this is mainly due to stronger optical confinement. Near $7.5 \mu\text{m}$, W_x is expected to pass through a minimum, then, together with that of TM_{10} and TM_{20} modes, increase gradually and strongly, as a result of weaker optical confinement in the direction of width. W_y , however, decreases further, and tends to become constant at larger strip widths. Similar behavior is observed for the wavelength $1.32 \mu\text{m}$ in the range from 5 to $10 \mu\text{m}$ by Korotky *et al.* [22]. It should be noted that the parameters are quite

different in wavelength and diffusion conditions between their experiment and this calculation, and it is not possible to give an accurate comparison for them. Nevertheless, previous calculated results indicated that plots of mode sizes for different wavelengths exhibit very similar characteristics [14], thus making it possible to perform a qualitative comparison as given above. The effective refractive indexes shown in Fig. 2(b) increase with the strip width W .

For quantitative comparison with published experimental data, the calculations of the variation of W_x and W_y with W were also performed on the basis of experimental data presented by Brinkmann [15] ($T = 1060 ^{\circ}\text{C}$, $H = 95 \text{ nm}$, $t = 9 \text{ h}$), and the computed results are shown in Fig. 2(c)

TABLE I

SINGLE-MODE, CUTOFF AND MULTIMODE CONDITIONS NEAR $1.53 \mu\text{m}$ FOR *c*-CUT Ti:LiNbO₃ WAVEGUIDES (*VALUES TAKEN FROM EXPERIMENTAL DATA IN [15])

	single-mode	cut-off	multimode	the other diffusion parameters
$W(\mu\text{m})$	3-7.5	below 3	above 7.5	$T=1050^\circ\text{C}$, $H=100 \text{ nm}$ and $t=8.3333 \text{ h}$
$W(\mu\text{m})$	3.5-8.5	below 3.5	above 8.5	$T=1060^\circ\text{C}$, $H=95 \text{ nm}$ and $t=9 \text{ h}^*$
$T(^{\circ}\text{C})$	above 1030	-	below 1030	$W=7 \mu\text{m}$, $H=100 \text{ nm}$ and $t=8.3333 \text{ h}$
$H(\text{nm})$	55-110	below 55	above 110	$W=7 \mu\text{m}$, $T=1050^\circ\text{C}$ and $t=8.3333 \text{ h}$

and (d). The single-mode region is for $W = 3.5 - 8.5 \mu\text{m}$, which is slightly different from the previous 3 to $7.5 \mu\text{m}$ because of the near diffusion parameters. This range includes the experimental data (6 and $7 \mu\text{m}$) reported by Brinkmann *et al.* [15].

C. Diffusion Temperature Dependence

In Fig. 3, we show the variations of W_x , W_y (TM₀₀ and TM₁₀) and corresponding N_{eff} with the diffusion temperature T for the case of $W = 7 \mu\text{m}$, $H = 100 \text{ nm}$ and $t = 8.3333 \text{ h}$. To facilitate the following discussion, the surface refractive index n_{es} versus T is also presented in Fig. 3(b) (n_{es} is independent of diffusion parameters and is held constant with a value of 2.1380). It is evident that the second mode is cutoff below the diffusion temperature 1030°C , in other words, when the diffusion temperature is above 1030°C , the waveguides support only fundamental mode TM₀₀. This result is supported by experimental data [15]; the $6 \mu\text{m}$ wide channel formed by indiffusing 95 nm thickness Ti-strip at 1060°C during 9 h is single-mode at wavelengths around $1.5 \mu\text{m}$. In Fig. 3(a) both W_x and W_y increase as the diffusion temperature is increased, which is proven by the near-field mode observed by Fukuma and Noda at the wavelength $1.15 \mu\text{m}$ [18]. In Fig. 3(b) the effective refractive indexes decrease with a rise in diffusion temperature.

D. Initial Ti-Strip Thickness Dependence

The dependence of mode sizes (TM₀₀ and TM₁₀) and corresponding effective refractive index on the initial Ti strip thickness H is plotted in Fig. 4. The other employed parameters are $T = 1050^\circ\text{C}$, $t = 8.3333 \text{ h}$ and $W = 7 \mu\text{m}$. Like Fig. 3(b), n_{es} versus H is also drawn in Fig. 4(b). The single-mode region is in the range from 55 to 110 nm , which contains the published experimental data $H = 95 \text{ nm}$ [15]. When H is below 55 nm , the mode is expected to cutoff, and above 110 nm the second mode TM₁₀ appears. In Fig. 4(a), both W_x and W_y decrease as the thickness is increased, and in the single-mode region, $W_x > W_y$. All these features agree qualitatively with the experimental results (at $1.32 \mu\text{m}$) reported by Korotky *et al.* [22].

E. Advantages of $1.48 \mu\text{m}$ Pumping

For Ti-diffused Er:LiNbO₃ waveguide devices, the preferred absorption band for optical pumping is around $\lambda_p = 1.48 \mu\text{m}$, at which both the ESA and optical damage of LiNbO₃ are weaker than pump wavelength in visible or near-infrared [10], [15]. Moreover, high-power laser diodes as pump sources

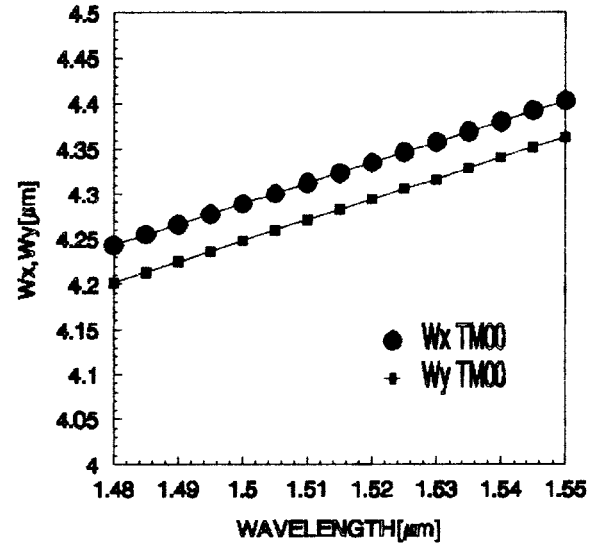


Fig. 5. Modes sizes versus wavelength for the case of $W = 7 \mu\text{m}$, $T = 1050^\circ\text{C}$, $H = 100 \text{ nm}$ and $t = 8.3333 \text{ h}$.

are commercially available at this wavelength. In addition, a previous study [14] has revealed that very little gaps between signal ($\lambda_s = 1.53 \mu\text{m}$) and $\lambda_p = 1.477 \mu\text{m}$ pumping mode sizes are found for the parameters W , T , and H , as a result of the little difference in wavelength. In Fig. 5 we show the variations of fundamental mode sizes with wavelength in the range from 1.48 to $1.55 \mu\text{m}$ (the other parameters are $W = 7 \mu\text{m}$, $T = 1050^\circ\text{C}$, $H = 100 \text{ nm}$ and $t = 8.3333 \text{ h}$). One can see that the dependence of W_x and W_y on the wavelength is slight, and the changes from 1.48 to $1.53 \mu\text{m}$ are merely $0.1 \mu\text{m}$, indicating that the single-mode condition are essentially identical for $1.53 \mu\text{m}$ and $1.48 \mu\text{m}$. Thus, the fabricated optical waveguides are simultaneously single-mode for both the pump and signal, which guarantees a good overlap of the two modes, and consequently a higher efficiency could be realized.

III. CONCLUSIONS

The dependences of the mode sizes and corresponding effective refractive indexes on controllable diffusion parameters at $1.53 \mu\text{m}$ wavelength were calculated for several lower order modes. The variational characteristics of mode sizes with these parameters have been outlined. When compared with published experimental results, a qualitative agreement has been obtained. The effective refractive index, as shown in Figs. 2–4, is dominated by the behavior of the mode sizes in appearance, and shows an inverse variable characteristic as compared to W_y . Based on the plots of both mode size

and effective refractive index, the single-mode and cutoff conditions are obtained and are summarized in Table I. Due to little wavelength difference between 1.53 and 1.48 μm the guide have the very closed single-mode and cutoff conditions, near-field distribution and mode sizes, and this assures a good mode overlap of signal and pump when Er:Ti:LiNbO₃ waveguide devices are pumped by $\lambda_p = 1.48 \mu\text{m}$ radiation, thus 1.48 μm is the preferred pump wavelength.

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Guilan Ding, photograph and biography not available at the time of publication.

Caihe Chen, photograph and biography not available at the time of publication.